Three lattice routes into ECDLP index calculus

Exploratory report, 2026-10-15. Idea Generator: `/root/lattice_exploration`, TASK-20261015-c848bd, RQ-IC-a39c7e. Input archive 266ad7929cca0cce1fca600543aab80c92107e27; handoff input revision 285ab8873505d6780d8490b19831b96caa2706d4. **These are proposals and review-pending derivations, not scientific findings.** No tests, timings, ECDLP recovery, hypothesis changes or approvals occurred.

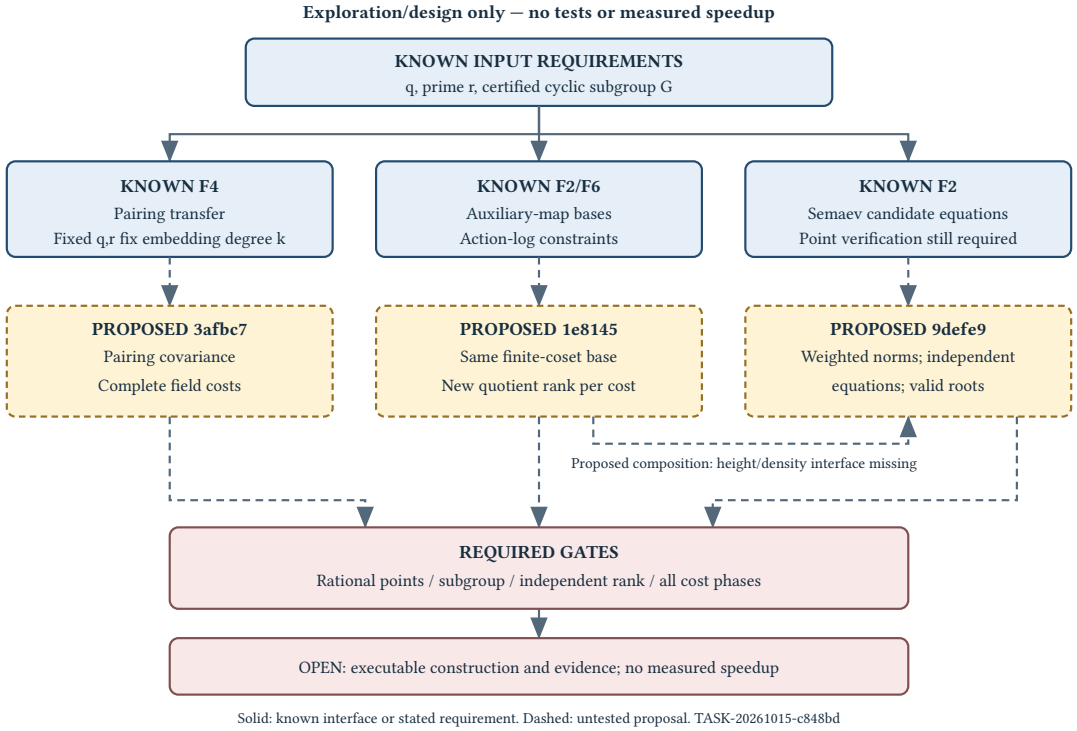


Figure 1: Known interfaces and proposed discriminations; dashed edges are untested candidate steps.

What each route would actually change

Proposal	Concrete mechanism	First useful discrimination	What remains unknown
IDEA-20261015-3afbc7	Compatible isogeny-adjoint pairing witnesses and explicit target-field bases	Verify tiny nondegenerate pairing covariance at unchanged q, r, k, then price map/pairing arithmetic	Complete field DLP/precomputation/descent costs; no easier field follows
IDEA-20261015-1e8145	Identical auxiliary finite-coset base as list, expanded map and low-degree chain	Exact rational base/density equality, followed by new quotient-rank per paid solve	Density, fresh-row rank yield, acquisition and solver cost
IDEA-20261015-9defe9	Dense versus Newton-target modular shifts with achieved norm certificates	Independent integer equations, root extraction and actual valid decomposition	Useful uniform-target coverage and unknown solver exponent σ

The user explicitly authorized F2/F4/F6 comparisons and compositions with F3 as an isogeny component. The proposals therefore retain known-family routes even when they supply no new lossy object. Each uses partial-action relation vectors; the pairing transfer itself is lossless, explicitly labeled as such. No family escape or general prime-field speedup is claimed. The three notes contain exact fixtures, controls, proof-search maps and literature boundaries.

Pairing: fixed field and subgroup fix the embedding degree

Write q for working field cardinality, r for the chosen prime subgroup order, and $k = \mathrm{ord}_r(q)$. A same-field isogeny of degree d coprime to r preserves that subgroup order and q, hence k. It may alter arithmetic presentation or point availability; it cannot lower k at those fixed inputs. This does not assert identical full rational point-group structures.

With a nondegenerate Weil witness S, pairing gives $h = e(Q,S) = e(P,S)^a = g^a$ for $Q = [a]P$. Under compatible transport of both arguments, $g' = g^d$ and $h' = h^d$. Relative to $\phi(P)$ the scalar is a directly. Check any alternative normalization separately. The point S, extension construction, map exceptions and nondegeneracy are inputs to certify and price, not free consequences of a small embedding degree. Root's Galbraith source scope covers the pairing/extension and isogeny context only, not an entire book or NFS audit. Galbraith, ch 26.3–26.4 and 25.4

The proposed unexecuted toy is $E/\mathbb{F}_{43} : y^2 = x^3 + x$, $E' : y^2 = x^3 - 4x$ with degree-two map $(x + 1/x, y(1 - 1/x^2))$, claimed count 44, $r = 11, k = 2$. Independently verify these claims, choose a certified order 11 point, and compare direct/transported/reused-power pairings over $\mathbb{F}_{43}[u]/(u^2+1)$. $S = (-x(P), u\,y(P))$ is a candidate distortion witness, not session evidence. Compare bases u and $v = u + 1$ with conversion charged. Degenerate $S = P$, a map that kills G, and a large-k gate are controls.

Powering by invertible d permutes μ_r. For uniform target images, a fixed descent cost function has the same subgroup average after that permutation. Selected-target variation alone is not an average descent improvement. A faster pairing presentation, alternative field solver or carefully priced reusable setup remains a legitimate separate comparison.

NFS uses a public ideal lattice; the ECDLP relation kernel is hidden

For $K = \mathbb{Q}(\theta)$ and a good prime ℓ with $f(s) = 0 \pmod{\ell}$, the special ideal $(\ell, \theta - s)$ imposes

$$a - bs = 0 \pmod{\ell}, \qquad L_{\ell,s} = \mathbb{Z}(\ell, 0) + \mathbb{Z}(s, 1), \qquad \det L_{\ell,s} = \ell.$$

This lattice is public before logarithms are known. Sieving organizes divisibility and smooth norms, followed by actual ideal/unit relation verification, filtering, LA and descent. The proposed tiny illustration uses $f_1 = z^2 + 1$, $f_2 = z^2 + 44$ modulo 43, first-side $(\ell, s) = (5, 2)$ and a fixed coefficient box; it is not optimized NFS or a recorded solve. A full performance proposal needs an exact primary NFS algorithm/regime and implementation beyond the supplied read scopes.

For ECDLP factor-base points F_i, the formal relation lattice $\{c : \sum c_i F_i = O\}$ has index r if they generate G. But its log-coordinate description $\sum c_i \log_P(F_i) = 0 \pmod{r}$ uses the unknown logarithms. Membership can be tested with group operations; a cheap short-vector basis is not thereby available. Its determinant is not a substitute for constructing relations. This is the missing interface in a direct NFS-lattice analogy.

Auxiliary chains: density and fresh rows, not raw orbit multiplication

PKM2016 already includes the auxiliary-map base primitive, including an auxiliary curve unrelated to the target. FGHR symmetry solving is also established in its stated extension-field setting; its abstract does not transfer a prime-field solver bound. The added question is the simultaneous density, membership, solver and new-rank cost on the **same actual base**. PKM2016, FGHR abstract

The proposed target is $E/\mathbb{F}_{97} : y^2 = x^3 + 2$ with candidate $\alpha(x, y) = (35x, y)$. Independently admit a prime cyclic subgroup and scalar action. Separately use $C_a : y^2 = x^3 + ax$ and the degree-two chain

$C_a \to C_{-4a}$ with $x \mapsto x + a/x$, $a_0 = 1$, $d = 0..3$. Choose a completely certified rational kernel and finite nonkernel coset $A + \ker \Phi_d$. Intersect its auxiliary x-values with actual target subgroup points. Compare list, expanded map and quadratic chain equations at identical base cardinality. Auxiliary rational lifts are not optional. No isogeny from this auxiliary curve to the target is asserted.

Finite-coset pole exclusions are justified because the chosen image is finite. Kernel/preimage-O bases require the projective/pole branches; generic denominator saturation can delete desired roots. A rank-two torsion negative fixture over $\mathbb{F}_{25}$ tests whether the endomorphism moves a chosen cyclic line. Without $\alpha(G) = G$, assigning a scalar λ is invalid.

For stable G and $\alpha(P) = [\lambda]P$, substituting $\log(\alpha^j F_i) = \lambda^j \log(F_i)$ makes every transformed copy of one augmented relation row a scalar multiple of the original. This does not deny solver sharing or independent rows from genuinely new decompositions. H-BINSTD-ce4f38 already studies this and is currently analyzed; its evidence was not audited here. IDEA-20260922-261004 is also prior rank work. The fact is not new.

Price all setup, membership, failed solves, orbit completion/canonicalization, filtering, reduced-rank checks, LA, descent and verification. If a raw base is orbit-completed, recount its larger actual M and quotient dimension U. Under a separately justified repeated-attempt model, rank-deficit stage j costs $C_j/(p_j\eta_j)$, where p_j is valid-decomposition probability and η_j conditional new-rank yield. Neither a constant yield nor an exponent is established.

Small roots: soundness, density and extraction are different gates

Use a coordinate window of width W but count actual $M = \mid F_W \mid$. With ordered m-tuples and repetition, $\sum_R N(R) = M^m$, so

$$\mathbb{E}_R N(R) = M^m/r, \qquad \Pr[N(R) > 0] \le \min(1, M^m/r).$$

This exact first moment is a necessary density bound. It does not provide a Poisson success law, independent rows or a cheap solver. Signs, unordered tuples, subgroup filtering and orbit duplicates change the appropriate counting unit.

The proposed lattice rows are $X^a f_R^b p^{t-b}$, $0 \le b \le t$. Compare ordinary total-degree shifts with exactly specified Newton-target support shifts. Scale coefficient columns by the root box. Let D count supported monomials and n be row rank; D need not equal n. For an achieved polynomial vector v, the strict certificate

$$D \parallel v \parallel_2 < p^{2t}$$

implies its value at every bounded modular root is zero over the integers. The weighted lattice's row covolume is $\sqrt{\det(BB^T)}$; only the square full-rank case permits $\mid \det B \mid$. No determinant-only LLL guarantee is asserted. These are explicit review-pending norm deductions, not a blanket multivariate Coppersmith theorem. Coppersmith's scoped abstract and Kosters–Yeo's abstract justify neither general root extraction nor a first-fall-degree runtime claim. Coppersmith abstract, Kosters–Yeo abstract

Many short coefficient-independent polynomials may have a common factor and a positive-dimensional zero set. Require actual independent integer equations, bounded-root extraction and verification of all original conditions. A Semaev zero is only a candidate: recover rational y-values, signs and subgroup membership, then verify the sum and new row rank. The geometric summation-polynomial framework is known; root read its author abstract/metadata, not its blocked PDF. Semaev

The proposed tiny grid is $(p, A, B) = (43, 1, 0), (97, 0, 2), (101, 2, 3)$, $W = 2, 4, 8$, $m = 2, 3, 5$, $t = 1, 2$, $L = 0, 1, 2$, with a common row-rank cap 64 and monomial cap 256. Full tiny oracle, planted/uniform targets, rational-lift and positive-dimensional controls are specified in the note. Overflow is an instrument limit, not mathematical failure. H-ECDLP-c3d48a covers only Stage 0 thresholds; H-ECDLP-

a00090 only a degree table. No small-root experiment is inferred from them. For univariate modular slices, compare direct finite-field factorization/root finding before claiming an LLL advantage; brute force is not the only applicable baseline.

In the restricted constant-yield fixed-m model $M = r^{\beta}$ and per-attempt r^{σ}, collection plus classical sparse LA is $r^{\beta+\sigma} + r^{2\beta}$. Subroot bookkeeping requires $\beta + \sigma < 1/2$, $\beta < 1/4$ and $\beta m \geq 1$, hence $m \geq 5$ only in that model. The exponent σ and every success/independence assumption remain unknown; construction, extraction, descent, memory and verification can dominate.

Low-degree auxiliary chains do not automatically compose with small-root methods: their intermediate field coordinates may require boxes near p. A useful mixed method needs a demonstrated density-preserving low-height/sparse encoding and paid search for small intermediates. Record this interface rather than combining two conditional advantages for free.

Source receipt and next action

All 51 rows in the two committed known-results renderings and the fresh dedup context were read. Only matching indexed designs were opened. The six primary retrieval scopes belong to `/root` in `inputs/source-read-receipts.json`:

Source	Root's inspected scope	Boundary retained here
Galbraith	ch 26.3–26.4 and 25.4 excerpts	No whole-book or NFS proof audit
PKM2016	abstract/introduction, 2.1, 3.1–3.4, Algorithm 1/partial cost	Existing primitive; no practical prime-field speedup
FGHR	author abstract/metadata	PDF 403; no full-paper inspection
Kosters–Yeo	version 3 abstract	No universal restricted-root impossibility
Coppersmith	IBM author abstract	No general modular multivariate guarantee
Semaev	author abstract/metadata	PDF 403; rational/subgroup checks still needed

No extra source inspection, historical evidence review or novelty certification is claimed. Root's working deductions were challenged by separating invariance from availability, row multiplicity from reduced rank, mean from success, and norm soundness from independent extraction. KB unavailable and experiment inventory unmaterialized do not imply absence. No canonical graph is changed: the supplied maps were checked as prior-art boundaries, and no new verified scientific edge exists.

Recommended first discrimination is the route 2 exact auxiliary/target base and subgroup admission check; it can cheaply expose empty density or invalid scalar action before a solver comparison. In parallel future design work, route 1 needs pairing covariance controls and route 3 needs achieved certificates plus independent root extraction. These are **proposed next tests**, each requiring its own executable protocol and canonical admission. The shared obstruction and exit criteria are in the problem record. Root will render and inspect the PDF in the separate visual task. All three routes remain open; no experimental result or status transition is requested by this report.